



3-D FE heat transfer simulation of quasi-simultaneous laser transmission welding of thermoplastics

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Abstract

Laser transmission welding of thermoplastics is a non-contact welding process for producing aesthetically and qualitatively high-grade joints with low thermal and mechanical stresses. In this research work, a three-dimensional heat transfer model is developed to simulate the quasi-simultaneous variant of laser transmission welding of thermoplastics. The thermal analysis is implemented to acquire the temperature history in this multi-pass welding process. The transient temperature field is computed using ANSYS[®] finite element code. The multi-pass movement of laser beam is realized by implementing a subroutine in ANSYS[®] Parametric Design Language (APDL). The effect of process variables namely laser power, welding speed, number of welding passes and line energy on temperature field during welding is investigated. The predicted outcomes of the temperature profiles, weld pool and time–temperature history are useful for optimizing the process parameters in the quasi-simultaneous laser transmission welding process.

Keywords Laser transmission welding · Quasi-simultaneous welding · Thermoplastics · Finite element model · Thermal analysis

1 Introduction

Laser transmission welding is a non-contact, non-contaminating, flexible joining process for thermoplastics that produces high-grade welded joints, optically and qualitatively [1]. In this process, two thermoplastic parts are welded by a laser beam, preferably in a lap joint configuration [2]. The thermoplastic part which is kept on the top should be either transparent or translucent to confirm that the significant proportion of the laser beam should penetrate that part and reach to the interface of the lap joint. The bottom part is an opaque thermoplastic which is laser absorbing. The application of laser in laser transmission welding is dependent on the thermo-optic interaction of beam with the thermoplastic parts. The beam energy absorbed by the bottom part is converted to heat energy, and both the plastic parts are melted at interface as and when the temperature reaches beyond the

glass transition or melting point of the thermoplastics used in joint assembly. After cooling and solidification, a firm and aesthetic weld seam is produced at joint interface [3, 4].

Laser transmission welding is now used in a wide range of application areas, including medical devices, automotive components, electrical and electronic devices, packaging, light and displays, house hold goods, and textiles industries [3]. Several applications are there in automotive industries for welding automotive parts such as connectors, front and rear lights assemblies, bumpers, liquid containers, dash board components, remote door keys, flood lights, and automotive intake manifolds. [5].

Laser transmission welding has four process variants, namely contour welding, simultaneous welding, quasi-simultaneous welding and mask welding [4]. In quasi-simultaneous laser transmission welding, the laser beam is moved rapidly over the welding path several times by a scanning galvo mirror system. Laser beam must be moved necessarily fast to avoid substantial cooling of the weld zone between two consecutive laser scans. The repeated laser exposures heat up the weld contour gradually and about evenly. Because of plastic's low thermal conductivity, the welding path is melted almost at the same time, as the name implies, quasi-simultaneously [6–8]. Figure 1 represents a

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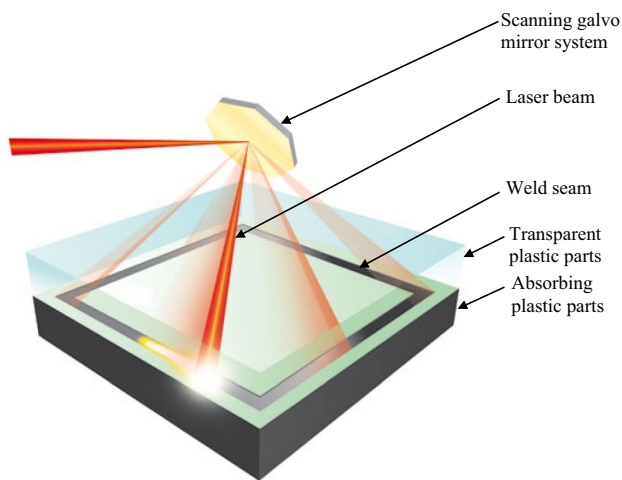


Fig. 1 Schematic view of quasi-simultaneous laser transmission welding process

schematic view of quasi-simultaneous laser transmission welding process.

Quasi-simultaneous technique has advantage of very short processing times that is why this technique is suitable for mass production. Quasi-simultaneous laser welding offers stronger welds than contour welding [7]. It is possible to bridge larger air gaps with quasi-simultaneous welding than with contour welding [7, 8]. However, the scanner work field sets the limits for weldable area of the part to be welded using this technique. The number of scans and the welding speed are the parameters that affect the welding time [9]. Both parameters affect the laser power required for adequate welding strength and joint quality [10]. Laser power, scanning speed and number of scans must be matched together to get good weld properties [9]. Pilli et al. [11] presented two-dimensional thermal model of quasi-simultaneous laser transmission welding of polymer. Computer software Flexi 2D is used for estimating weld geometry and degradation rate of the weld. Potente et al. [12] studied the parametric effects on weld quality during quasi-simultaneous welding of poly-ether-ether-ketone (PEEK) using an Nd:YAG laser. Jansson et al. [13] examined the parametric effects on contour laser welding as well as quasi-simultaneous laser transmission welding of thermoplastics. For this, laser power, welding speed and beam spot size are varied for contour welding, whereas an additional parameter as number of scans is considered for quasi-simultaneous welding. Jansson et al. [8] studied the quasi-simultaneous welding of polypropylene with a fiber coupled diode laser. Air gaps are bridged up to 30 μm . Ruotsalainen et al. [9] studied quasi-simultaneous laser welding of transparent polymers using a fiber laser. A new scanning technique, the beam path off-setting, is applied during scanning of weld path. Schmailzl et al. [14] proposed a technique of monitoring

weld seam during quasi-simultaneous laser welding, which is very first laser scanning technique, by using a pyrometer incorporated with a 3-D scanner. This scanning method is further used to observe gap-bridging in a T-joint configuration, during quasi-simultaneous laser welding [15]. Ghasemi et al. [16] investigated the meltdown phenomena during quasi-simultaneous laser welding of polycarbonate and polypropylene and correlated the degree of meltdown with different process parameters, viz. laser power, welding speed, welding pressure, number of scans and scan length. Mamuschkin et al. [17] investigated the process of selective laser energy deposition during quasi-simultaneous laser welding of absorber-free plastics and achieved to minimize the heat affected zone. Though more than a few experimental investigations on quasi-simultaneous laser transmission welding is carried out, the numerical modeling of quasi-simultaneous laser transmission welding is rare. Numerical modeling of the welding processes provides a deeper insight into the process mechanisms, while at the same time reduces experimentation costs and efforts. A three-dimensional heat transfer model is capable of predicting the thermal field along the weld line as well as along the transverse direction and along the direction of weld penetration. Thus, it is required to develop a three-dimensional numerical model of quasi-simultaneous laser transmission welding for process simulation, thermal analysis, parametric investigation and optimization of the process.

In this paper, a three-dimensional thermal model of quasi-simultaneous laser transmission welding process is developed. The model is used to study the transient heat transfer phenomena during the quasi-simultaneous welding process. Effect of welding parameters on weld pool temperature is also investigated. The model is able to predict the time-dependent temperature history in three-dimensional space, which can further be used to predict the weld seam dimensions and for process optimization. In addition to thermal analysis, the developed model is extended to a sequential thermo-mechanical model, where the temperature results of thermal model are subsequently used in mechanical analysis as inputs to determine thermally induced weld distortion.

2 Finite element modeling

This section presents the details about the numerical modeling of quasi-simultaneous laser transmission welding process. A transient three-dimensional heat transfer model is developed to predict the temperature distribution within and around the weld pool in three-dimensional space. A sequentially coupled field model is developed by utilizing the output of thermal model as input in a mechanical model. This mechanical model is utilized to study the evolution of welding distortion in laser transmission welding process.

2.1 Materials and methods

Polycarbonate is used as the work material. The top part of the assembly is a transparent polycarbonate plaque, whereas the bottom part is a laser absorbing polycarbonate plaque which is rendered black by adding 0.2% carbon black by weight. Quasi-simultaneous variant of laser transmission welding is used for welding the thermoplastic parts. Diode laser of 809.4 nm wavelength is used to perform the weld. The laser energy distribution follows Gaussian distribution. The energy delivered by the laser beam is largely governed by conduction into the surrounding material of the thermoplastic parts, for the duration of operation, and a small amount of heat is lost by convection and radiation from the exposed surfaces of the parts.

2.2 Transient heat transfer model

The generation of the transient temperature field while laser welding is simulated based on the heat conduction mechanism:

$$\rho c \left(\frac{\delta T}{\delta t} \right) = \{L\}^T ([D] \{L\} T) + q_v \quad (1)$$

where ρ is the density of the material (kg/m^3), c is the specific heat (J/kg K), T is the location and time-dependent temperature (K), t is the time (s), $\{L\}$ is the vector operator, and $[D]$ is the conductivity matrix

$$\{L\} = \begin{Bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{Bmatrix}; \quad [D] = \begin{bmatrix} k_{xx} & 0 & 0 \\ 0 & k_{yy} & 0 \\ 0 & 0 & k_{zz} \end{bmatrix} \quad (2)$$

where K_{xx} , K_{yy} , and K_{zz} are the thermal conductivity (W/m.K) in the element along x , y and z direction, respectively. The term q_v in Eq. (1) is the rate of internal heat generation per unit volume (W/m^3). To calculate the internal heat generation rate (q_v) at different nodes, Lambert Beer's equation [17], which indicates that the intensity of the monochromatic radiation beam in the absorbing medium declines exponentially with the depth of penetration, is utilized with a Gauss expression for Gaussian laser intensity distribution [18],

$$q_v = \begin{cases} 0 & \text{for transparent part} \\ \{T_t(1 - R_a)\} \frac{P}{\pi r_0^2} \exp\left(-\frac{r^2}{r_0^2}\right) & \text{for absorbing part} \\ \exp(-Kz_a) & \end{cases} \quad (3)$$

where T_t is the transmissivity of the transparent polymer part and R_a is the reflectivity of absorbing polymer. Thus, the expression $T_t(1 - R_a)$ is used to determine the portion of laser beam enters to the bulk of the absorbing material. Other variables of Eq. (3) namely P is optical power (W), K is the coefficient of absorption (m^{-1}) and z_a is the depth from interface (m) for the absorbing material, r_0 is radius of the laser beam, r is radial distance of a point on the material's surface at interface; $r = \sqrt{(x_s^2 + y_s^2)}$, where x_s and y_s are the Cartesian coordinates of that point.

Heat loss due to free convection and radiation from the surfaces exposed to the surroundings are considered as boundary conditions. Specified convection and radiation surfaces acting over surface S are given by:

$$\{q\}^T \{\eta\} = h(T_s - T_B) + \sigma \varepsilon (T_s^4 - T_B^4) \quad (4)$$

where $\{\eta\}$ is the unit outward normal vector, h is the coefficient of convective heat transfer ($\text{W/m}^2 \text{K}$), T_s is the surface temperature (K) of the model, T_B is bulk temperature (K) of the adjacent fluid, ε is emissivity, and σ is the Stefan Boltzmann constant ($5.67 \times 10^{-8} \text{ W/m}^2 \text{K}^4$). Convective heat transfer coefficient, h , is assumed to be $5 \text{ W/m}^2 \text{K}$ [19]. Emissivity of polycarbonate is taken as 0.95 [20].

At initial condition (at $t = 0 \text{ s}$), the bulk temperature of the workpieces is assumed to have uniform room temperature. The bulk temperature of the adjacent fluid is chosen as $T_0 = 27^\circ \text{C}$ (300 K).

2.3 Plasticity model

The nonlinearity of materials is triggered by the nonlinear relationship of stress and strain, i.e., stress is a nonlinear function of strain. The total strain vector in the laser transmission welding problem can be described as:

$$\{\varepsilon\} = \{\varepsilon^{\text{el}}\} + \{\varepsilon^{\text{pl}}\} + \{\varepsilon^{\text{th}}\} \quad (5)$$

where $\{\varepsilon\}$ is total strain vector, $\{\varepsilon^{\text{el}}\}$ is elastic strain vector, $\{\varepsilon^{\text{pl}}\}$ is plastic strain vector, and $\{\varepsilon^{\text{th}}\}$ is thermal strain vector.

The stress is related to the strain by:

$$\{\varepsilon\} = D^{-1} \{\sigma\} \quad (6)$$

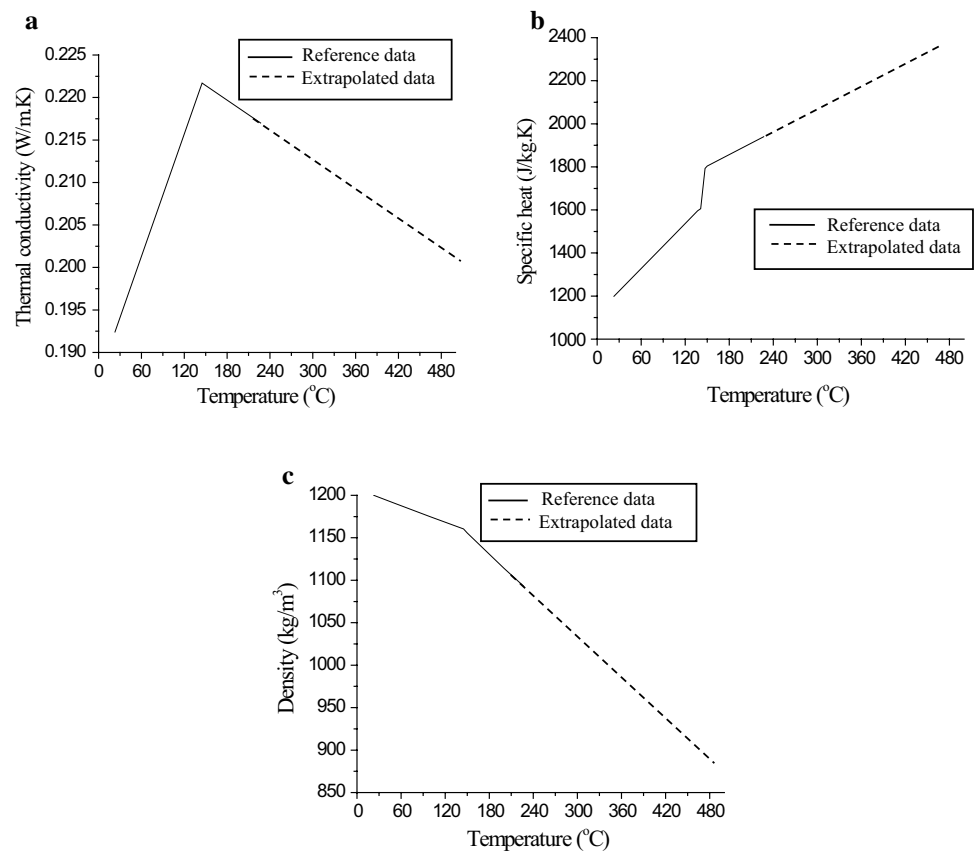
where $\{\sigma\}$ is the stress vector, D is stiffness matrix, and $\{\varepsilon\}$ is total strain vector.

The compliance matrix D^{-1} is:

$$D^{-1} = \begin{bmatrix} 1/E & -\nu/E & 0 \\ -\nu/E & 1/E & 0 \\ 0 & 0 & 1/G \end{bmatrix} \quad (7)$$

where E is Young's modulus, ν is Poisson's ratio, and G is the shear modulus.

Fig. 2 Temperature dependent **a** thermal conductivities, **b** specific heat, and **c** density of polycarbonate



2.4 Thermo-physical, optical and mechanical properties of polycarbonate

The temperature response in a material involving high heat fluxes is determined by the temperature-dependent thermo-physical properties, namely thermal conductivity, specific heat and density. The thermal conductivity of polycarbonate as a function of temperature is shown in Fig. 2a [21, 22]. Figure 2b presents the temperature-dependent specific heat of polycarbonate [22]. The density of polycarbonate dependent on temperature is taken from open literature [21, 22] and shown in Fig. 2c.

The optical properties of polymers (transmission, reflectance and absorbance) are one of the most important parameters required for modeling the laser transmission welding process, because it determines to what extent and where the laser energy in the polymer is absorbed. Optical properties of the polymer are wavelength dependent. Kagan et al. [23] reported that highly transmittable polymers (in their natural state) are not sensitive to wavelength changes in the range from 400 to 1080 nm. Transmittance of polycarbonate plaques is measured using a Gentec-EO SOLO-2 laser power and energy meter. Transmittance is measured using the power readings from the power meter before and after hitting the transparent part. For the transparent polycarbonate plaque of 2 mm thickness, transmittance of $93 \pm 1\%$ (at 809.4 nm wavelength of diode

laser) is obtained. The reflectance has two contributions—specular (from the front and rear surfaces of the specimen) and diffuse (from irregular surfaces and within the bulk of the specimen). For optically clear polymers, only specular reflection is significant and the reflectance can be calculated from the reflection loss [23]. Siegel and Howell [24] described the relationship between the specular reflectance (η_i) from an

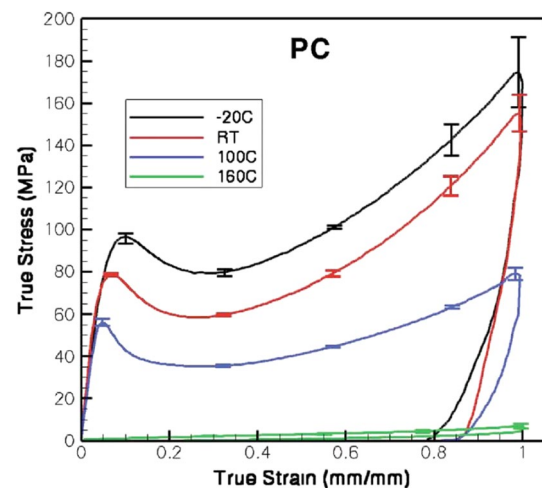
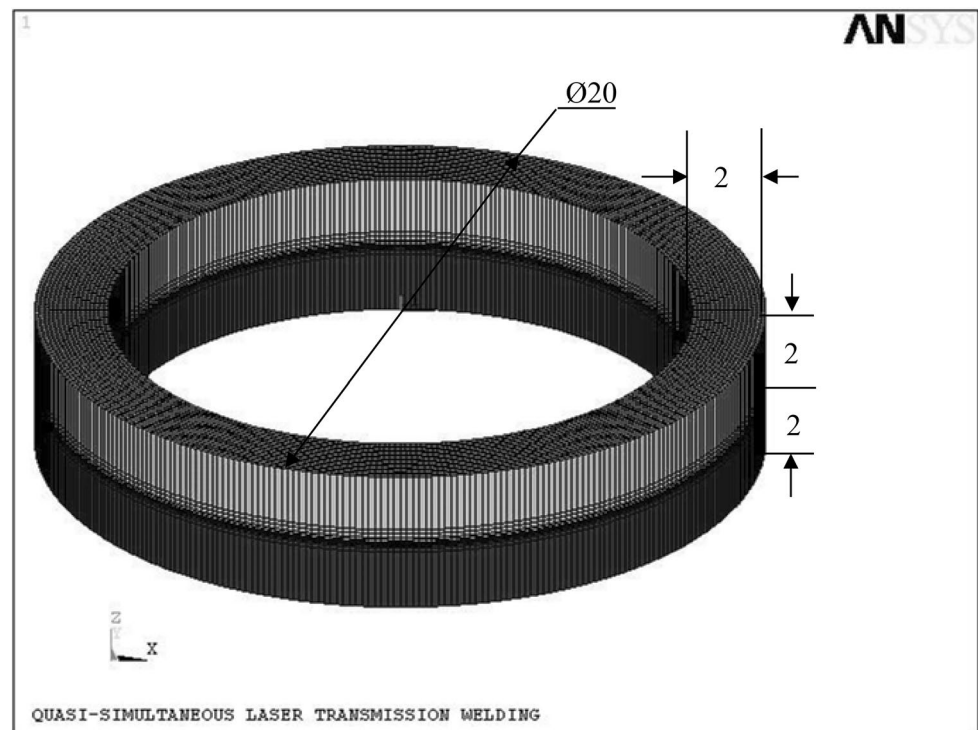


Fig. 3 True stress–strain curves of polycarbonate at different temperatures

Fig. 4 Physical configuration used for simulation of quasi-simultaneous welding process



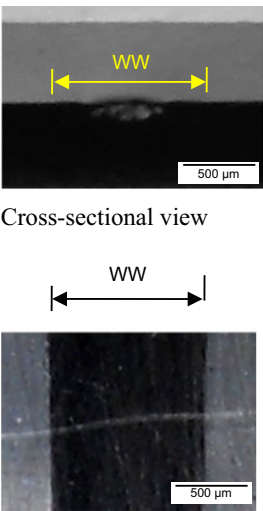
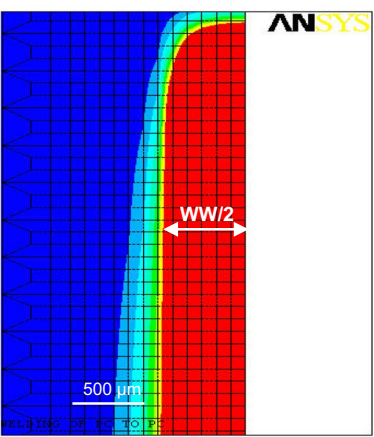
Welding parameters: Power: 12 W Welding speed: 15 mm/s Beam diameter: 1 mm Carbon black content of absorbing polymer: 0.2 %wt. Weld width Predicted: 1.14 mm Measured: 1.0246 mm	 Cross-sectional view Top view	 Top view (weld interface)
WW = weld width WW/2 = half weld width	Experimentally measured weld width	Predicted half weld width (half of the model using symmetry)

Fig. 5 Comparison of the FE model predicted weld width result with the experimentally measured result

interface between two materials having indices of refraction, n_1 (for polymer) and n_0 (for air)

$$\eta_i = \left(\frac{n_1 - n_0}{n_1 + n_0} \right)^2 \quad (8)$$

Given the refractive indices for polycarbonate ($n_1 = 1.58$) and air ($n_0 = 1$), reflectance of polycarbonate at an air-polymer interface is expected to be 5.2%. However, it varies with wavelength. Kagan et al. [25] reported that polycarbonates only specularly reflect about 7% of the incident beam

intensity, at the wavelength of diode and Nd:YAG lasers. Russek et al. [26] measured the optical penetration depth of polycarbonate containing 0.2 wt% and 0.3 wt% carbon black. Optical penetration depths of 60 μm and 71 μm are obtained at 808 nm and 940 nm laser wavelengths, respectively, for polycarbonate containing 0.2 wt% carbon black. For polycarbonate containing 0.3 wt% carbon black, optical penetration depths of 55 μm are obtained at the 808 nm laser wavelength. Absorption coefficients can be calculated simply from the optical penetration depths as it is reciprocal of this.

The stress–strain behavior of the material above the yield point is of prime importance for elasto-plastic analysis. For elasto-plastic analysis, the FEM requires besides modulus of elasticity and Poisson's ratio also input of a uniaxial true stress–strain function. Modulus of elasticity, also known as elastic modulus or Young's modulus, is a measure of how a material will deform and strain when placed under stress.

The Young's modulus of polycarbonate at ambient temperature is 2.4 GPa [21]. A much lower value of the Young's modulus ($=5\text{ MPa}$) is considered above 150 $^{\circ}\text{C}$ (glass transition temperature of polycarbonate) to describe softening of the material above this temperature. The Poisson's ratio is the proportion of the transverse contraction strain to the longitudinal expansion strain in the pulling force direction. Poisson's ratio of polycarbonate is 0.38 [21]. For nonlinear assessment, true stress and true strain are needed. The engineering stress–strain does not correctly depict the physics when the strains become large. The sample length rises during pulling, but the cross section shrinks. In engineering stress/strain, the assumption is that there is little change in the original cross-sectional region. This only applies to small strain, especially in the elastic area. Once yielding occurs, the values of stress and strain must be based on the instantaneous area of the cross section and length. The true stress at any load is the load divided by the cross section at

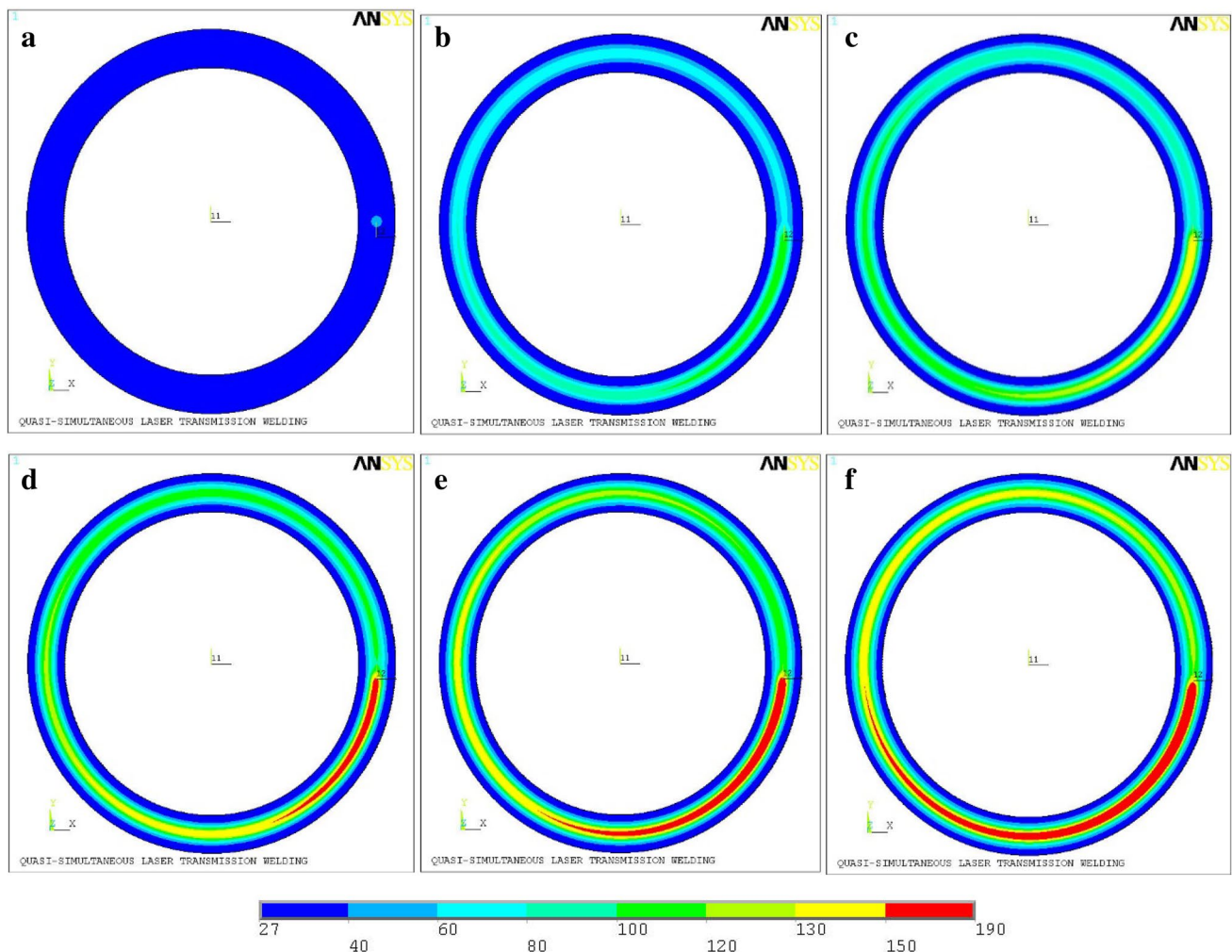


Fig. 6 Distribution of temperature at weld interface during different instances: **a** when the laser beam approaches the parts, **b** after 1 scan, **c** after 2 scans, **d** after 3 scans, **e** after 4 scans, and **f** after 5 scans

(power=20 W, welding speed=0.5 m/s, beam diameter=1 mm and number of scan=5)

that time. Figure 3 shows the true stress–strain curves of polycarbonate [27].

2.5 Modeling using ANSYS

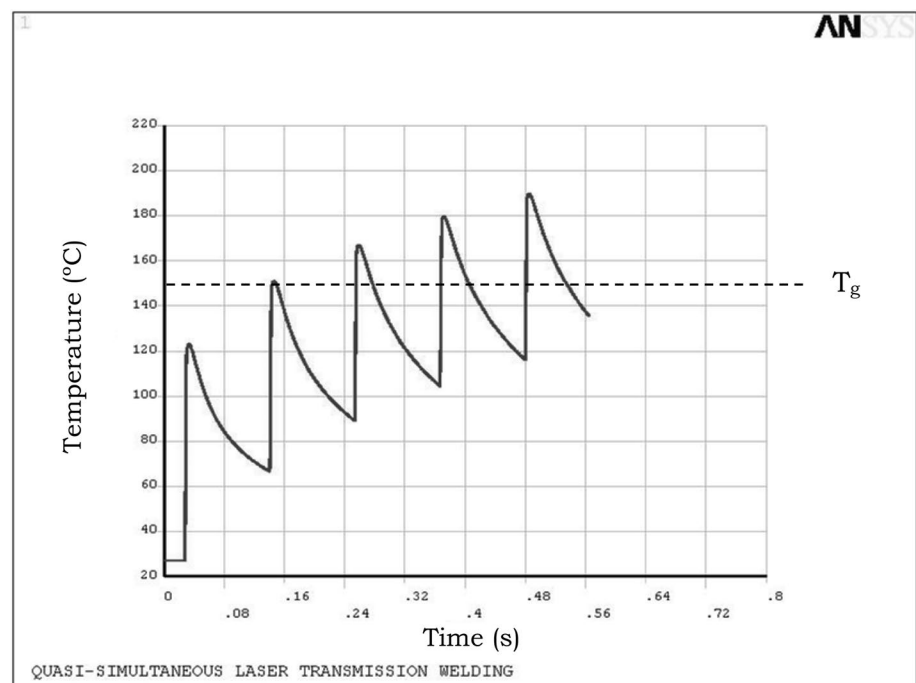
The physical configuration used for simulation of quasi-simultaneous welding process is shown in Fig. 4. The top part is transparent polycarbonate, while the bottom is an opaque polycarbonate plaque. This geometry is considered for reducing the number of nodes to reduce the solution time, without compromising the accuracy of the results (which is sensitive to node density at the irradiation zone). The length of the weld path is 57 mm. The model has total 85,680 nodes. Running the model for a single scan (360 load steps) takes approximately 3.5 h of solution time with a computer configuration of 2.66 GHz Intel Core™ 2 Duo processor and 3 GB RAM. The time required for solving the model for multiple scans is generally couple of hours to couple of days, depending on the number of scans. However, a higher end computer configuration can reduce the simulation time drastically. It requires nearly 7 GB disk space for storing load step result files for a single scan.

Thermal brick element, SOLID70 [28] is used to mesh the whole model. SOLID70 is capable of 3-D thermal conduction. A 3-D steady-state or transient thermal analysis is applicable to the element. A varying density mesh pattern is created by reducing the total number of nodes in order to minimize simulation time. Because of the high heat flux in this part, the region near the laser irradiation area, fine

meshes are created. A perfect contact between mating parts and an isotropic thermal behavior of the work material is considered as simplification assumptions for development of the FE model. The moving heat source due to laser radiation is modeled based on ANSYS® Parametric Design Language (APDL). The position of the beam center (x, y) is moved forward in each load step by increasing the value of x and y by $x = D \cos \theta$ and $y = D \sin \theta$, where D is the distance from the beam center to the plate center, and θ is the angular increase in the position of the laser beam. In this routine, small angular increments are imposed to heat loads using *DO loop command. For realizing a multi-scan laser movement, this routine is placed inside another *DO loop command. The outer loop executes the number of scans and the inner loop executes circular movement of the beam for single scan.

The plasticity model used in the present research used a multi-linear isotropic hardening type curve (MISO) for implementing the true stress–strain data. With the related flow rule and isotropic hardening, this model utilizes the von Mises yield criterion. For plasticity analysis, the thermal element, SOLID70, is replaced by an equivalent structural element, SOLID45 [28]. Temperatures results extracted from thermal analysis is subsequently applied as body loads on elements at the nodes. During coupled field analysis, only one quarter of the 3-D model is used for minimizing the simulation time by considering symmetry of the model (axisymmetric model). Thereby, the number of load steps is reduced, and the time steps are calculated accordingly. Running full 3-D model for thermo-mechanical

Fig. 7 Time–temperature history during the welding process (power = 20 W, welding speed = 0.5 m/s, beam diameter = 1 mm and number of scans = 5)



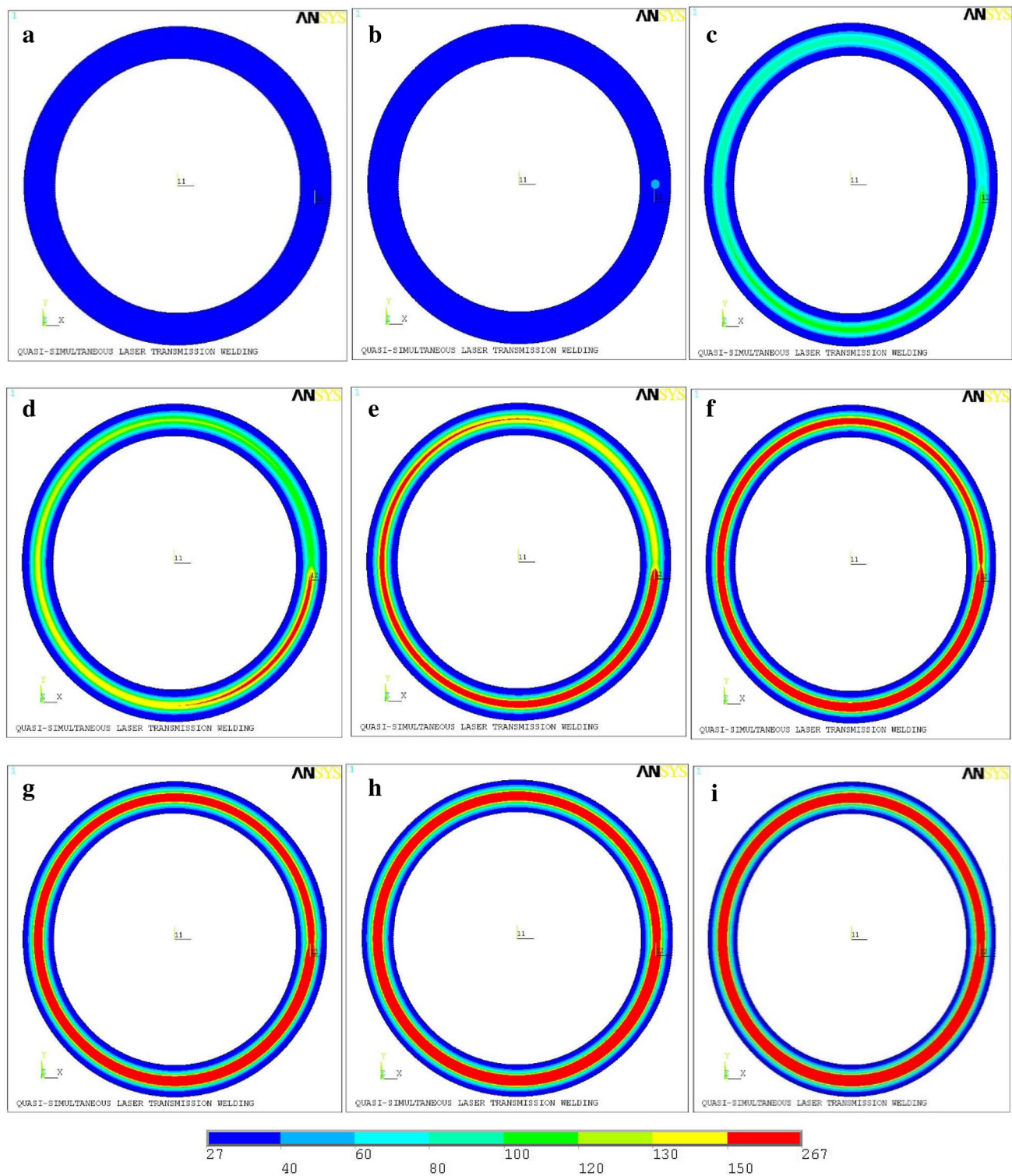


Fig. 8 Distribution of temperature at weld interface: **a** before welding, **b** when the laser beam approaches the parts, **c** after 1 scan, **d** after 2 scans, **e** after 3 scans, **f** after 4 scans, **g** after 5 scans, **h** after 6

scans, and **i** after 7 scans (power = 40 W, welding speed = 1 m/s, beam diameter = 1 mm and number of scans = 7)

sequential coupled analysis with all the load steps and for multiple scans takes couple of days to reach to the solution. However, for the thermal analysis the full model is used for simulation.

2.6 Validation of the model

The model is validated indirectly due to the unavailability of experimental setup for quasi-simultaneous laser welding. The available contour laser welding (welding in straight line) setup is used for model validation. For this, a 3-D FE model of laser transmission contour welding is developed first and validated the model prediction using experimental result, by utilizing the available contour laser welding setup. Figure 5 illustrates the comparison of results obtained through laser transmission contour welding of polycarbonates through experiments and FE modeling. The red region at the weld interface denotes fusion zone where temperature exceeds the glass transition temperature ($150\text{ }^{\circ}\text{C}$) of polycarbonate. The comparison shows that they agree fairly, thus validating the physics of model. Then, to develop the quasi-simultaneous laser welding model, the same physics (heat source, boundary conditions, optical and thermo-physical properties of materials, and simplification assumptions) is used. It is therefore expected that the results of the developed FE model will be consistent based on the known material data.

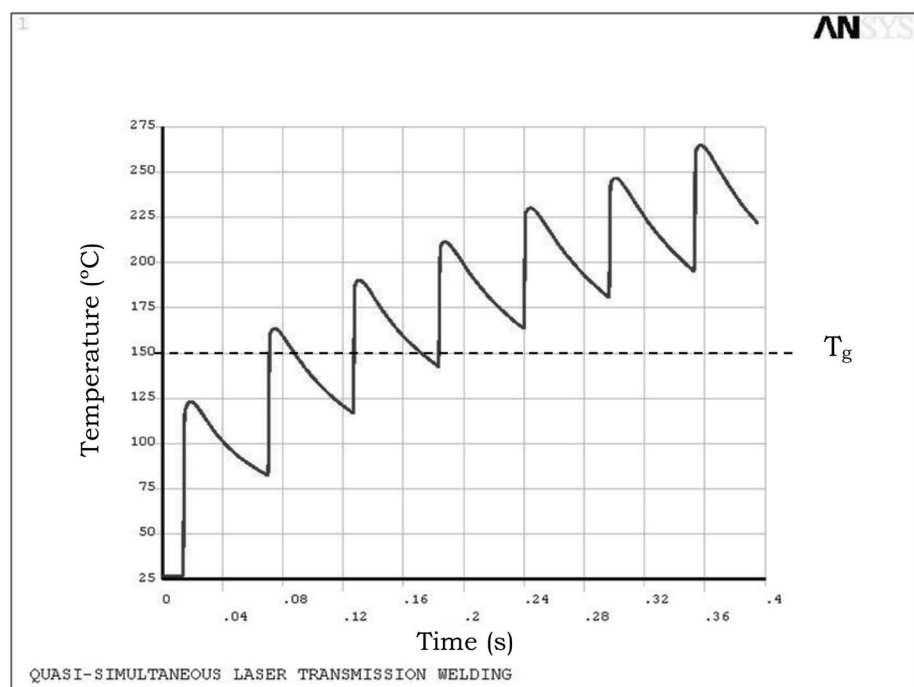
3 Result and discussion

Figure 6 displays the temperature distribution in different instances at the weld interface, which is computed for 20 W laser power, 0.5 m/s welding speed and 1 mm beam diameter.

Number of scans used is 5. From this figure, it is seen that the temperature rises gradually with the number of scans and reaches to the glass transition temperature of polycarbonate ($150\text{ }^{\circ}\text{C}$) during 3rd scan. After 5 number of scans, the maximum weld zone temperature reaches to $190\text{ }^{\circ}\text{C}$. Figure 7 shows the temperature history during the welding process. It is depicted from above figure that for every weld scan, the temperature at the selected point at weld seam rises rapidly when the laser beam approaches that point. Because of plastic's low thermal conductivity and very small-time interval between two consecutive scans, the temperature drop is less compared to temperature rise, during each scan. Due to accumulation of these residual temperatures, the temperature increases with the number of scans. The higher the number of scans during the welding process, the lower the temperature rise and fall.

Figure 8 displays the distribution of temperature in different instances at the weld interface, which is computed for 40 W laser power, 1 m/s welding speed and 1 mm beam diameter. Number of scans used, in this case, is 7. It can be seen from the figure above that the welding path (seam) begins to melt almost simultaneously, i.e., quasi-simultaneously, when the number of scans is 7. The quasi-simultaneous melting of the weld path is dependent on the laser power, welding speed and number of scans used. It

Fig. 9 Time–temperature history during the welding process (power = 40 W, welding speed = 1 m/s, beam diameter = 1 mm and number of scans = 7)



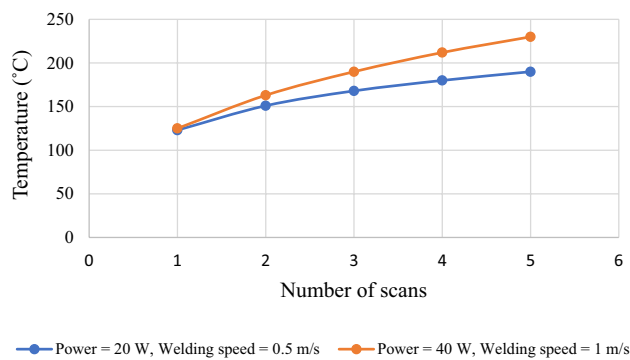


Fig. 10 Effect of line energy on maximum weld temperature

is also observed from this figure that with higher number of scans wider welds can be achieved.

It is observed from Figs. 7 and 9 that the weld zone temperature reaches to a maximum of 190 °C after 5 scans when the applied power and welding speed are 20 W and 0.5 m/s, respectively, whereas after same number of scans the maximum temperature of 231 °C is obtained for a power and welding speed of 40 W and 1 m/s, respectively. In both the cases, the line energy, i.e., the ratio of power to welding speed, is same, i.e., 40 J/m. Though the energy input per unit length is same for both the cases, a combination of higher power with higher welding speed results higher temperature at weld zone than that for a

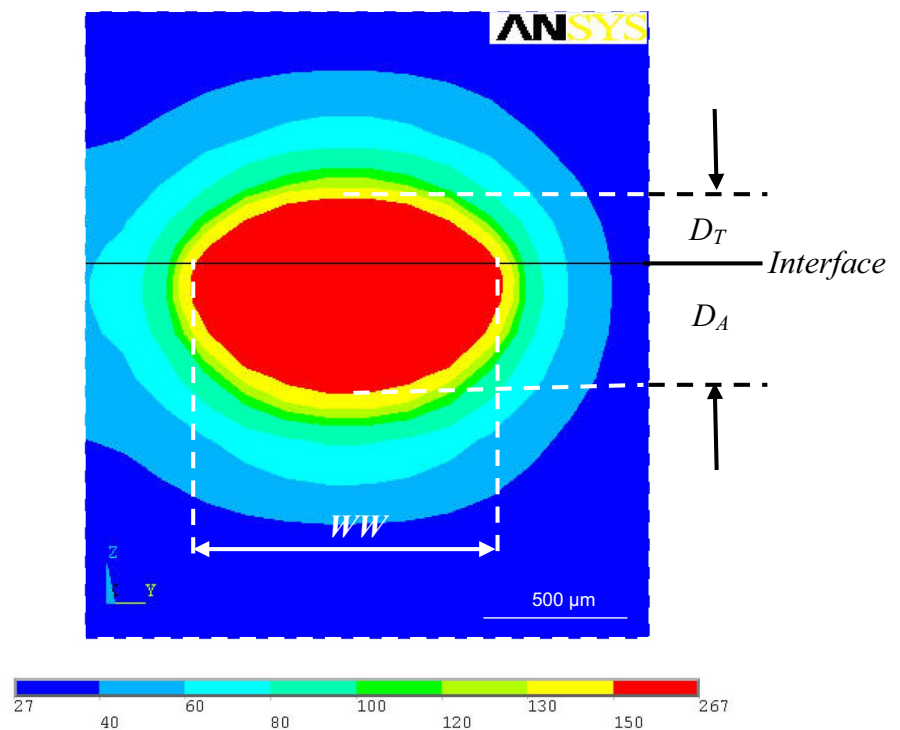
combination of lower power with lower welding speed, which is illustrated in Fig. 10.

The relatively low laser power requires low welding speed to maintain required line energy, and thus heat transfer comes into play a major role here, because heat conduction losses are more pronounced at these slow speeds. Use of higher laser power allows the use of higher welding speed to attain the same line energy, which causes less time available for heat dissipation, and thus maximum of the input energy is deposited at the weld zone.

The process parameter combination should be carefully selected to obtain a desired weld. As a 3-D model, the present model can also predict the weld seam width and depth of penetration within the transparent and absorbing materials, as shown in Fig. 11.

The effects of individual process parameter on weld quality, quantified as weld dimensions, are studied using scatter plots presented in Figs. 12, 13 and 14. It is observed from Fig. 12 that all the weld dimensions (weld width and depth of penetration within transparent and absorbing part) are increasing with laser power. As laser power increases at a fixed welding speed (welding speed = 1 m/s, number of scans = 5, beam diameter = 1 mm), more base material is melted, and the weld seam becomes wider owing to a shift in the distribution of intensity in the target area. From Fig. 13 it is noted that the welding speed has negative impact on all weld dimensions. With increasing welding speed at a fixed laser power (laser power = 30 W, number of scans = 5, beam diameter = 1 mm),

Fig. 11 Prediction of weld width (WW) and depth of penetration within the transparent part (D_T) and absorbing part (D_A)



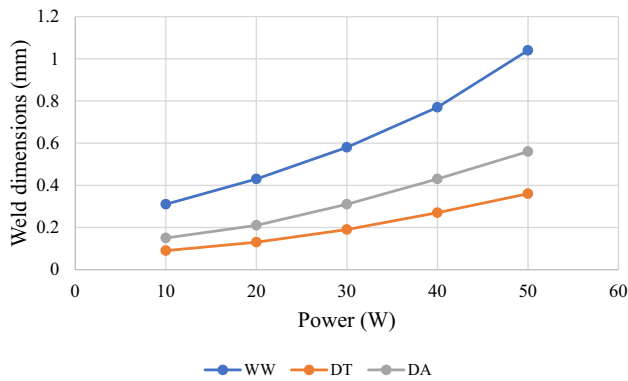


Fig. 12 Effect of power on weld dimensions (WW=weld width, DT and DA=depth of penetration within the transparent part and absorbing part, respectively)

owing to shorter laser–material interaction time and reduced energy per unit length, the weld seam becomes narrower. It is seen from Fig. 14 that all weld dimensions are increasing with number of scans. Due to the fast and repetitive thermal cycles during laser scanning in quasi-simultaneous welding, increasing number of scans when all other parameters are fixed (laser power=30 W, welding speed=1 m/s, beam diameter=1 mm), increases the melt volume at weld, resulting in increasing of weld dimensions.

Welding process parameters influence the degree of distortion mainly through their effect on the heat input. Welding distortion is the result of the introduction, redistribution or relaxation of residual stresses during and after welding process. Thermal distortion predicted by the model is given in Fig. 15. The maximum distortion (distortion vector sum, UEQV in meter) observed after completion of welding process is of the order of $6.6 \mu\text{m}$ (for power=40 W, welding speed=1 m/s, beam diameter=1 mm, number of scans=7, clamping pressure=1 MPa). The accumulation of plastic strain is responsible for permanent weld distortion. The plastic strain remains as residual strain after welding. Plastic strain develops when and as the applied load exceeds the yield strength of the material. The yield stress of the material becomes very low after reaching melting/softening temperature. Because of this, a part of the specimen at and near the bead deforms plastically under relatively lower load which causes weld distortion.

4 Conclusion

The transient temperature field of quasi-simultaneous laser welding process is simulated using ANSYS® finite element code. This developed model can be adapted to a variety of plastic materials, geometry, optical properties and boundary conditions. During thermal analysis using the developed

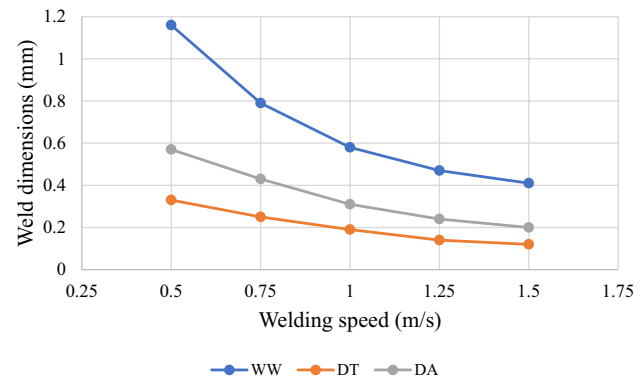


Fig. 13 Effect of welding speed on weld dimensions (WW=weld width, DT and DA=depth of penetration within the transparent part and absorbing part, respectively)

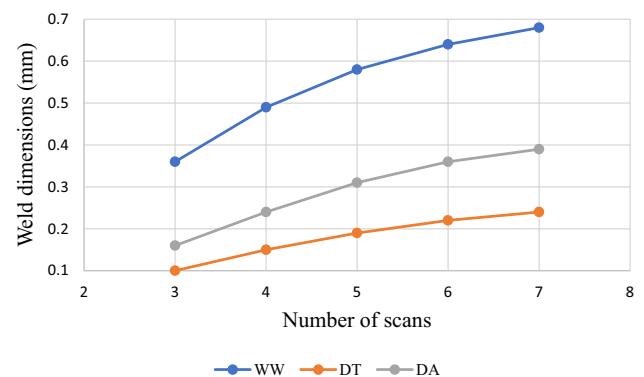
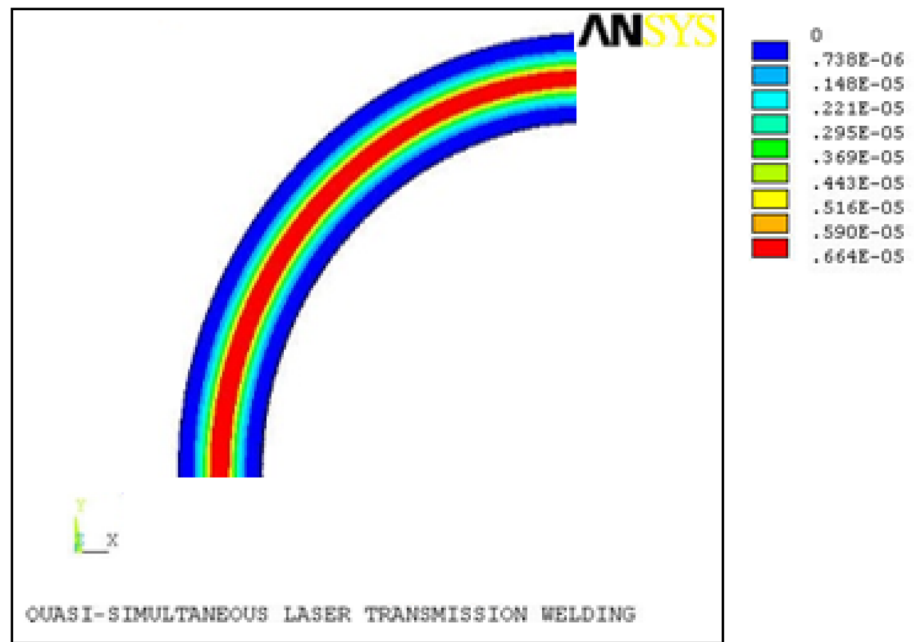


Fig. 14 Effect of number of scans on weld dimensions (WW=weld width, DT and DA=depth of penetration within the transparent part and absorbing part, respectively)

model, it is observed that the weld zone temperature, during each scan, rises rapidly, but temperature drop is comparatively less because of low thermal conductivity of plastics and very small-time interval between scans. This results in accumulation of residual temperatures, which increases with number of scans, and causes melting of entire weld zone quasi-simultaneously. It is further seen that the quasi-simultaneous melting of weld zone depends on combination of laser power, welding speed and number of scans used as they depict the laser energy deposition and laser–material interaction time. For same energy input per unit length, a combination of higher power with higher welding speed results higher temperature at weld zone. It is also observed that weld width increases with laser power and number of scans, whereas decreases with the increase of welding speed. The resulting weld distortion observed after completion of welding process is of the order of $6.6 \mu\text{m}$. The developed 3-D model would be used for prediction of optimal process window for quasi-simultaneous laser welding in future study of this work.

Fig. 15 Axisymmetric contour obtained for weld distortion vector sum, UEQV (m)



Compliance with ethical standards

Conflict of interest The author declares that he has no conflict of interest.

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